

HIV Antiviral Agents

Daniel Anderson, PharmD, BCIDP • 34-page lecture deck • life-cycle target → drug class → toxicity → clinical use

Core Lecture 8

ART goals

NRTI / NNRTI / INSTI / PI

Entry + capsid inhibitors

PrEP / PEP / U=U

SOURCE PRIORITY: the attached Anderson *Antiretroviral Agents* deck is primary. The attached high-yield handout is secondary enrichment. Its heading says “Lecture 10,” but for the current Week 8 course organization this resource is Core Lecture 8: HIV Antiviral Agents, per your correction.

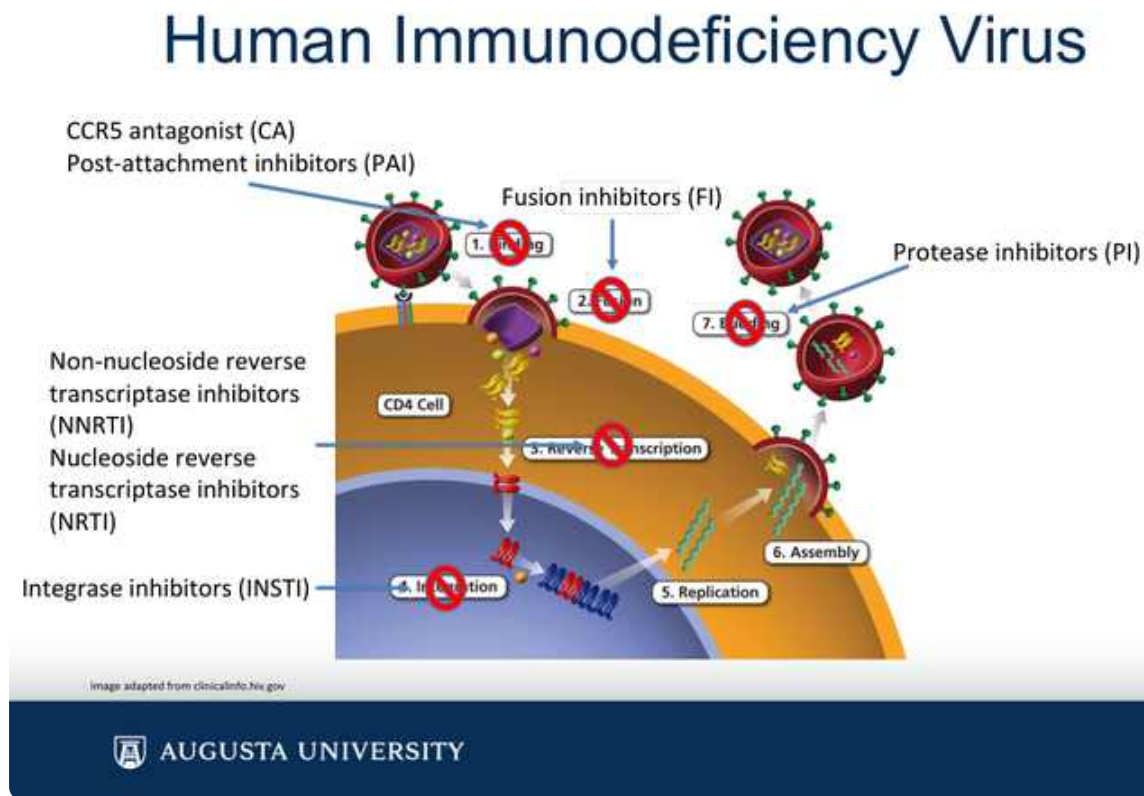
Lecture Objectives

1. Explain the goals of antiretroviral therapy (ART).
2. Describe the mechanisms of action of antiretrovirals.
3. Recognize the most common adverse effects of ART.
4. Apply PrEP, PEP, treatment as prevention, rapid initiation, and Test & Treat to patient care.

HOW TO STUDY THIS LECTURE

Do not memorize an alphabet soup of drugs in isolation. Start with the HIV life-cycle step, attach the drug class to that step, then learn the few toxicity / interaction clues the deck actually highlights.

Slide 4: The Whole Lecture on One Diagram



Slide 4. The lecture maps the major antiretroviral classes directly onto the HIV life cycle: entry/fusion, reverse transcription, integration, and maturation.

Life-cycle step	Class / agent	What is blocked
Attachment / coreceptor use	CCR5 antagonist; attachment/post-attachment agents	HIV interaction with CD4/coreceptor machinery
Fusion	Fusion inhibitor class	Fusion of viral and host membranes
Reverse transcription	NRTI + NNRTI	Viral RNA → DNA
Integration	INSTI	Insertion of HIV DNA into host genome
Maturation	Protease inhibitor	Cleavage of viral polyproteins into functional proteins
Multiple capsid-dependent steps	Lenacapavir	Nuclear transport, viral assembly, capsid assembly

Ozzy checkpoint - click him once you can point to the HIV life-cycle step before naming the drug class.

Handout-only mechanism detail: the companion handout explains NRTIs as chain terminators, NNRTIs as allosteric reverse-transcriptase inhibitors, and gives enfuvirtide as the gp41 fusion-inhibitor example. Use those to deepen the mechanism, but the current slide deck's class map is the source of truth for what is emphasized.

Slides 5-8: What ART Is Trying to Accomplish

Continuum of Care

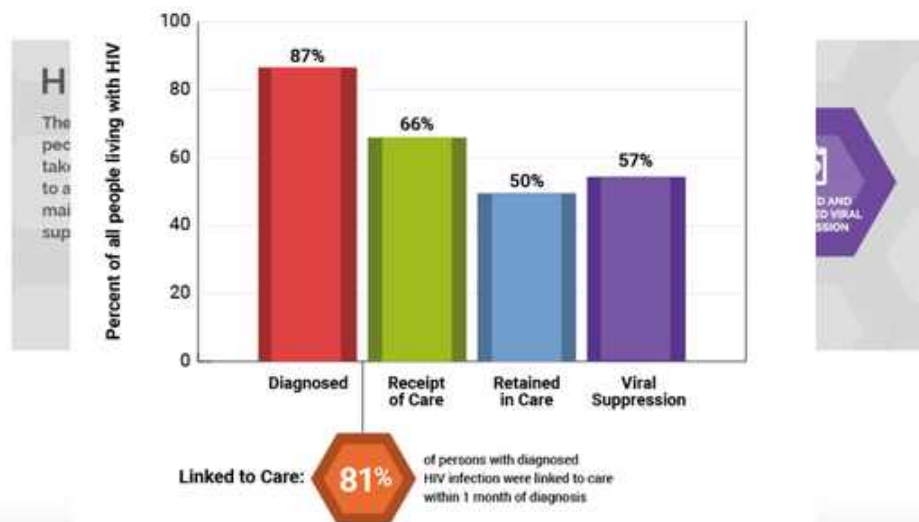


Image adapted from hiv.gov



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Slide 5. The continuum-of-care slide frames HIV treatment as more than prescribing a drug: diagnosis, linkage to care, retention, and viral suppression all matter.

Goals of Antiretroviral Therapy (ART)

Virologic Suppression

- Undetectable viral load
- No replication = no mutation
- **U = U**

Immunologic

- CD4 cell count recovery
- Reduced risk of opportunistic infections

Clinical

- Prolong life expectancy

Societal/Financial

- Reduce spread of HIV
- Reduced personal, institutional, and societal costs associated with HIV

Slide 6. The deck organizes ART goals into virologic, immunologic, clinical, and societal/financial outcomes.

ART GOALS

Virologic: undetectable viral load; no replication → less mutation; **U = U**.

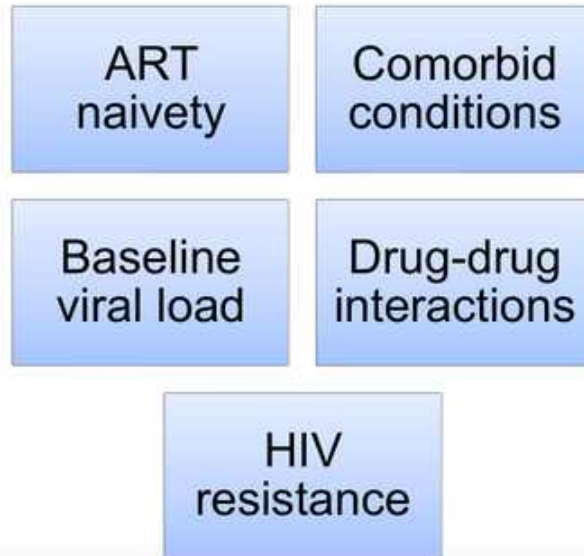
Immunologic: CD4 recovery and fewer opportunistic infections.

Clinical: prolong life expectancy.

Societal: reduce HIV transmission and costs.

Deck timing: virologic suppression generally occurs within the first **3-6 months** of ART. Predictors include lower baseline viremia, potent therapy, tolerability/convenience, and adherence.

Considerations for Starting ART



Slide 8. Starting ART is individualized around ART history, comorbid conditions, baseline viral load, drug-drug interactions, and HIV resistance.

Slides 9-10: NRTIs - The Backbone Class

Nucleoside Reverse Transcriptase Inhibitors (NRTIs)

- Historically, considered the “backbone” of therapy

Currently Used In Practice	No Longer Recommended
Abacavir Emtricitabine Lamivudine Tenofovir disoproxil fumarate Tenofovir alafenamide Zidovudine - neonates	Stavudine Didanosine Zidovudine - adults

Older NRTI Side Effects
Mitochondrial toxicity Myopathy Hepatic steatosis Lactic acidosis Lipoatrophy Bone marrow suppression Peripheral neuropathy

Slide 9. The deck distinguishes currently used NRTIs from older agents and emphasizes that many of the classic mitochondrial toxicities belong primarily to older drugs.

Drug	Current-deck pearl	Major adverse effect / consideration
TDF	Nucleotide; commonly paired with emtricitabine	Nephrotoxicity / Fanconi syndrome; decreased bone mineral density
TAF	Nucleotide; commonly paired with emtricitabine	Same toxicity pattern as TDF but less severe in the deck
Abacavir	Nucleoside; paired with lamivudine	HLA-B*5701 hypersensitivity ; cardiovascular events; never rechallenge after rash; no HBV activity
Emtricitabine	Nucleoside; paired with tenofovir	Well tolerated
Lamivudine	Nucleoside; paired with abacavir	Well tolerated
Zidovudine	Older NRTI	Older class toxicities; still used for perinatal prophylaxis

NRTI Pharmacology

Drug	Structure	Combined With	Side Effects	Other Considerations
Tenofovir disoproxil fumarate (TDF)	NucleoTide	Emtricitabine	- Nephrotoxicity (Fanconi syndrome) - Decreased bone mineral density	
Tenofovir alafenamide (TAF)	NucleoTide	Emtricitabine	Same as TDF, but less severe	
Abacavir	Nucleoside	Lamivudine	- Hypersensitivity – HLA-B*5701 - Cardiovascular events	- Never re-challenge rash - Not active against hepatitis B
Emtricitabine	Nucleoside	Tenofovir	Well tolerated	
Lamivudine	Nucleoside	Abacavir	Well tolerated	
Zidovudine	Nucleoside		See older ART side effects table	Used for perinatal prophylaxis

Slide 10. The pharmacology table makes TDF vs TAF and abacavir's HLA-B*5701 hypersensitivity the major practical distinctions.

ABACAVIR MUST-NOT-MISS

HLA-B*5701 positive → do not use abacavir. If hypersensitivity/rash occurs, the deck says **never rechallenge**.

Handout-only mechanism detail: NRTIs are phosphorylated by host kinases and act as faulty nucleoside/nucleotide building blocks that terminate the growing viral DNA chain because they lack the required 3'-OH. The handout also supplies additional specific pearls such as zidovudine macrocytic anemia/bone-marrow suppression.

Slides 11-12: NNRTIs - Watch the Rash and Interactions

Non-Nucleoside Reverse Transcriptase Inhibitors (NNRTIs)

Drug	Initial Therapy	Food Requirement	Side Effects	CYP Drug-Drug Interactions	Other Drug Interactions
Doravirine	Yes	No	Well tolerated	3A4 substrate	No
Rilpivirine	Yes*	With meal	- Depression - Headache - Rash - QTc prolongation	3A4 substrate	Acid suppressors
Efavirenz	No	Empty stomach	- CNS (dizziness, insomnia, vivid dreams) - Lipodystrophy - Hyperlipidemia	3A4 substrate 2B6 substrate - Inhibitor of 1A2, 2C8, 2C9, and 2C19 - Inducer of 2A6	
Nevirapine	No	No	- Rash - Lipodystrophy	3A4 substrate 2B6 substrate - Inducer of 3A4 and 2B6	No

Slide 11. The deck compares doravirine, rilpivirine, efavirenz, and nevirapine with their food, adverse-effect, CYP, and drug-interaction patterns.

Drug	Deck clue
Doravirine	Well tolerated; CYP3A4 substrate; no food requirement
Rilpivirine	Take with a meal; depression/headache/rash/QTc prolongation; acid suppressors interact
Efavirenz	Empty stomach; CNS effects including dizziness, insomnia, vivid dreams; lipodystrophy/hyperlipidemia
Nevirapine	Rash and lipodystrophy; CYP interactions

NNTRI Class Side Effects

Skin Rash

Stevens
Johnson
Syndrome

Hepatitis

Low Barrier
to
Resistance*

Slide 12. NNRTI class warnings: skin rash, Stevens-Johnson syndrome, hepatitis, and a relatively low barrier to resistance.

Rilpivirine vignette: if a patient starts a strong acid-suppressing regimen and later loses virologic control, think **impaired rilpivirine absorption**, not automatically resistance.

Handout-only mechanism detail: NNRTIs bind an allosteric site on reverse transcriptase rather than mimicking nucleosides.

Slides 13-14: INSTIs - Modern Workhorse Class

Integrase Strand Transfer Inhibitors (INSTIs)

- Inhibits insertion of HIV-1 DNA into host genome. Creates circular non-integrated episomes instead
- **Generally, best tolerated, most efficacious class**
- **High barrier to resistance**

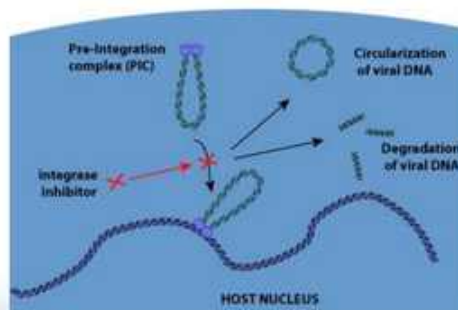


Image adapted from viralzone.org



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Slide 13. INSTIs prevent insertion of HIV DNA into the host genome. The deck calls them generally the best-tolerated, most efficacious class with a high barrier to resistance.

INSTI Pharmacology

Drug	Side Effects*	CYP Interactions	Other Interactions**	Miscellaneous
Bictegravir	- Nausea/diarrhea - Headache - CPK increase	3A4 substrate	- P-gp substrate - UGT1A1 substrate - OCT2 inhibition (metformin)	Decreased tubular secretion of creatinine without glomerular function changes
Dolutegravir	- CPK – myositis - Headache - Hepatotoxicity	3A4 substrate		
Elvitegravir	- Diarrhea - Headache	3A4 substrate • Co-formulated with 3A4 inhibitor		- Take with food - Requires a booster
Raltegravir	- Headache - Nausea	None	• UGT1A1 substrate	
Cabotegravir	- Injection site pain - Headache - Nausea	None	• UGT1A1 substrate	• Long-acting injectable

*Class side effects: **weight gain**, insomnia, depression

**Avoid polyvalent cations - chelation



Slide 14. INSTI pharmacology: bictegravir, dolutegravir, elvitegravir, raltegravir, and cabotegravir. Polyvalent-cation chelation is a class interaction to remember.

Drug	Deck pearl
Bictegravir	Nausea/diarrhea, headache, CPK increase; decreased tubular creatinine secretion without changing glomerular function
Dolutegravir	CPK/myositis, headache, hepatotoxicity
Elvitegravir	Take with food; requires a booster
Raltegravir	Headache/nausea; no CYP interaction highlighted
Cabotegravir	Long-acting injectable; injection-site pain/headache/nausea

INSTI CLASS

High barrier + well tolerated. Remember **polyvalent cations** → **chelation** and the deck's class effects of weight gain, insomnia, and depression.

Slide 15: Protease Inhibitors - Maturation + CYP

Protease Inhibitors (PIs)

- Mechanism of action: inhibits cleaving of polypeptide products of HIV mRNA into their functional parts
 - Prevents maturation of new virus
- All are metabolized by the liver (3A4) and **must be given with a pharmacological booster** (CYP inhibitor)
- Class side effects: hyperglycemia, GI intolerance, **lipodystrophy** (Cushing-like)

PIs		Boosters
Atazanavir	Lopinavir	Ritonavir
Darunavir	Nelfinavir	Cobicistat
Fosamprenavir	Ritonavir	
Indinavir	Saquinavir	

Slide 15. PIs block cleavage of HIV polypeptide products, preventing maturation of new virions. The deck emphasizes CYP3A4 metabolism, pharmacologic boosting, and metabolic adverse effects.

PI CORE

Protease inhibitor → immature/nonfunctional virions.

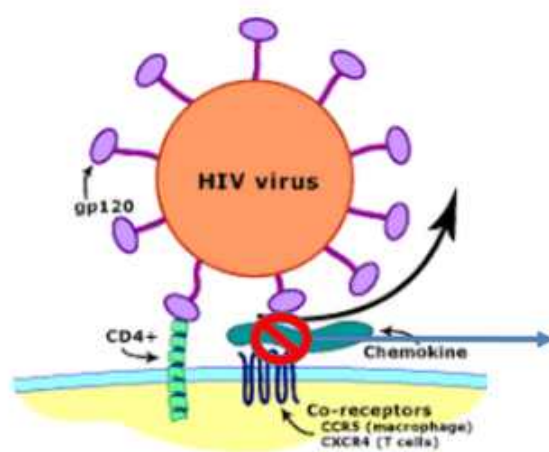
Ritonavir or cobicistat = pharmacologic boosters.

Class effects: **hyperglycemia + GI intolerance + lipodystrophy.**

Handout-only board pearls: the companion adds atazanavir → indirect hyperbilirubinemia and darunavir → sulfa-moiety caution. Those are useful board associations but are not the main content on slide 15.

Slides 16-19: Entry and Capsid Inhibitors

Coreceptor (CCR5) Antagonist - Maraviroc



- Requires viral tropism test
- Only works for CCR5 coreceptor binding
- Primary side effect: upper respiratory tract infections

Prevents gp120 and gp41 conformational changes to stop membrane fusion

Image adapted from thebodyprn.com



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Slide 16. Maraviroc blocks CCR5-dependent entry. It requires a viral tropism test and only works for CCR5-tropic virus.

MARAVIROC

CCR5 antagonist → **requires tropism testing.** If the virus uses CXCR4, maraviroc will not solve the entry problem.

CD4 Post-Attachment Inhibitor - Ibalizumab

- Humanized monoclonal antibody
- Binds to D2 complex of CD4 cell preventing conformational changes to gp120 complex
- Administered IV
- Side effects: GI, rash, hypersensitivity reactions

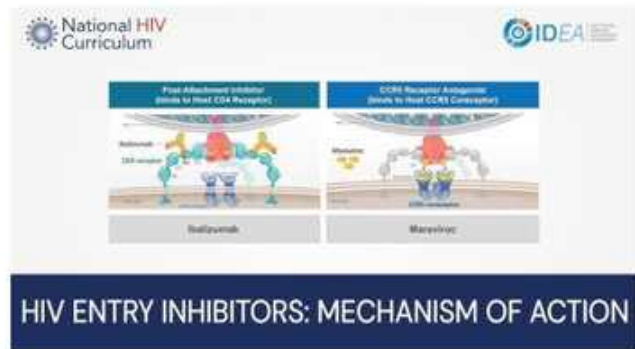


Image adapted from hiv.uw.edu

Slide 17. Ibalizumab is a humanized monoclonal antibody that binds the D2 region of CD4 and prevents post-attachment conformational changes.

Gp120 Attachment Inhibitor - Fostemsavir

- Side effects:
 - Nausea
 - Liver function test elevations
 - Sleep disturbances
 - QTc prolongation

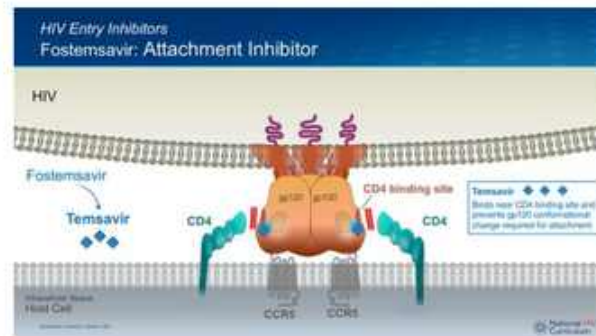


Image adapted from hw.uw.edu

Slide 18. Fostemsavir is presented as a gp120 attachment inhibitor; the slide emphasizes nausea, LFT elevations, sleep disturbance, and QTc prolongation.

Capsid Inhibitor - Lenacapavir

- Targets 3 distinct areas of HIV lifecycle
 - Nuclear transport
 - Viral assembly
 - Capsid assembly
- Treatment or prevention
- Oral or IM administration
- Side effects: injection site reactions, nausea, headache

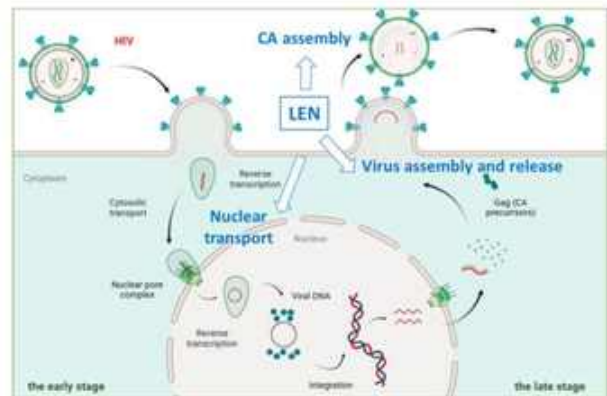


image adapted from sciencedirect.com

Slide 19. Lenacapavir is a capsid inhibitor targeting nuclear transport, viral assembly, and capsid assembly; the deck notes treatment or prevention and oral or IM administration.

Handout-only entry-agent addition: enfuvirtide binds gp41 and blocks membrane fusion; it is given subcutaneously and is associated with injection-site reactions.

Slides 20-22: Choosing a Regimen

Considerations

Medication Adherence

- Side effect profile
- One tablet, once a day regimen
- Tablet size
- Long-acting injectables?

Comorbidities

- Drug-drug interactions
- Side effect profile
- Pregnancy/breastfeeding safety

Baseline Viral Information

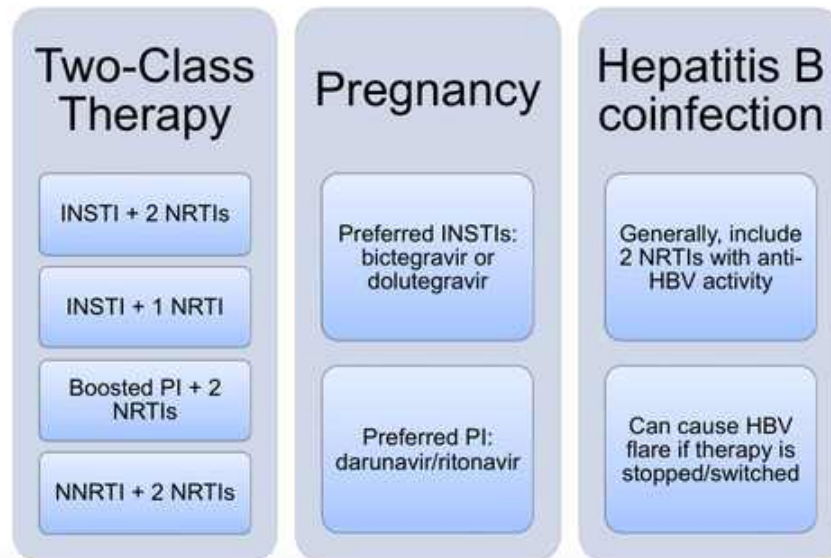
- Initial viral load
- Initial CD4 cell count



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Slide 21. Regimen choice is driven by adherence, comorbidities/drug interactions, pregnancy/breastfeeding considerations, and baseline viral information.

What to Start



Slide 22. The current deck lists several two-class treatment frameworks and then separately highlights pregnancy and hepatitis B coinfection considerations.

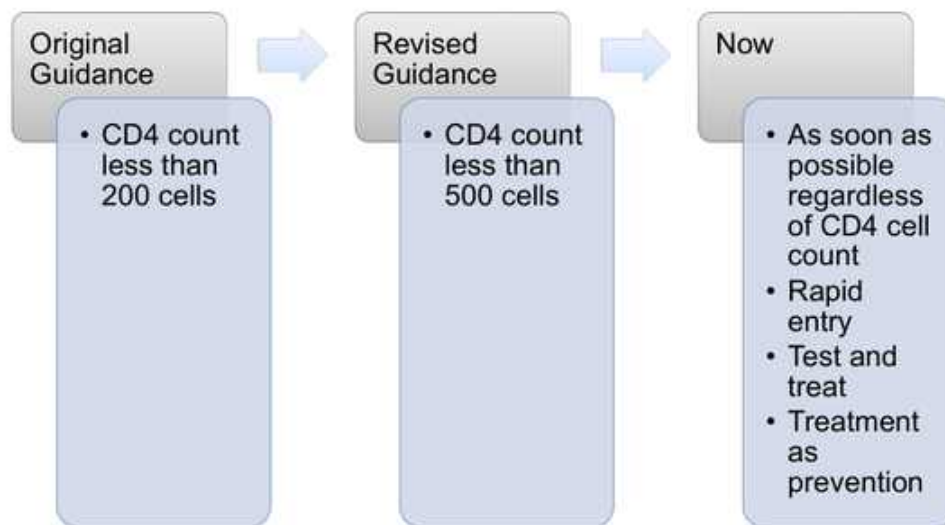
Current-deck regimen framework	What the slide says
INSTI + 2 NRTIs	Listed two-class therapy option
INSTI + 1 NRTI	Also listed as an option
Boosted PI + 2 NRTIs	Listed alternative framework
NNRTI + 2 NRTIs	Listed alternative framework
Pregnancy	Slide names bictegravir or dolutegravir as preferred INSTIs and darunavir/ritonavir as preferred PI
HBV coinfection	Generally include 2 NRTIs with anti-HBV activity; stopping/switching can trigger HBV flare

Deck vs handout mismatch: the older handout summarizes initial ART as “2 NRTIs + 1 INSTI” and says dual therapy should generally not be used initially. The **current**

lecture slide explicitly includes INSTI + 1 NRTI among its treatment frameworks, so this companion follows the deck rather than silently importing the older simplification.

Slides 23-25: Start Early, but Know the Exceptions

When to Start



Slide 23. The lecture shows the historical progression from CD4-based thresholds to starting ART as soon as possible regardless of CD4 count: rapid entry, Test & Treat, and treatment as prevention.

WHEN TO START

Now: as soon as possible regardless of CD4 count.

Keywords: rapid entry • Test & Treat • treatment as prevention.

Exceptions to Rapid Start

Cryptococcal
meningitis

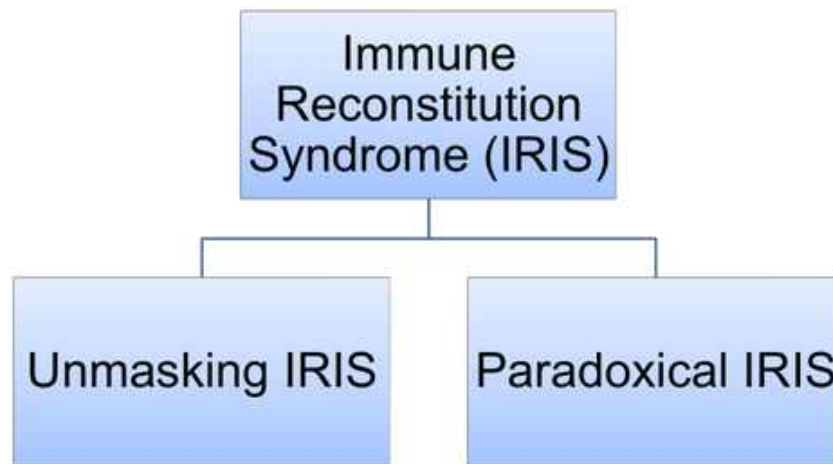
Tuberculosis
meningitis

Slide 24. The deck's rapid-start exceptions are cryptococcal meningitis and tuberculosis meningitis.

RAPID-START EXCEPTIONS ON THIS DECK

Cryptococcal meningitis and **tuberculosis meningitis**.

ART Initiation Complications



Slide 25. ART initiation can trigger immune reconstitution inflammatory syndrome (IRIS), divided here into unmasking and paradoxical IRIS.

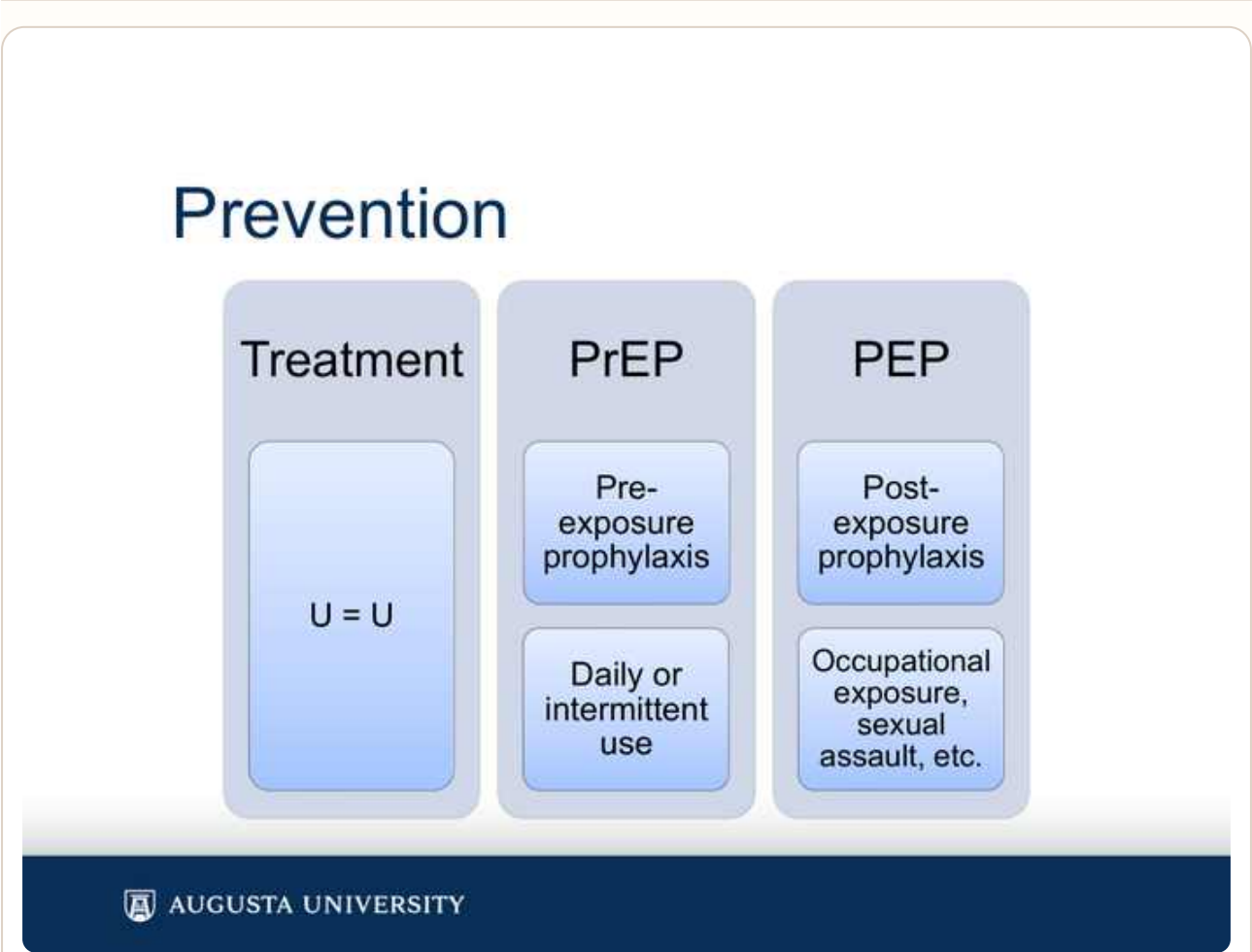
Unmasking IRIS

Previously unrecognized infection becomes clinically apparent as immune function returns.

Paradoxical IRIS

A known/treated infection clinically worsens after immune recovery.

Slides 26-27: Prevention - Treatment, PrEP, and PEP



Slide 26. The prevention slide organizes HIV prevention around effective treatment/U=U, PrEP before exposure, and PEP after exposure.

Concept	Use
U = U / treatment as prevention	Virologic suppression reduces sexual transmission risk
PrEP	Pre-exposure prophylaxis; daily or intermittent use is shown on the slide
PEP	Post-exposure prophylaxis after occupational exposure, sexual assault, etc.

PrEP Efficacy

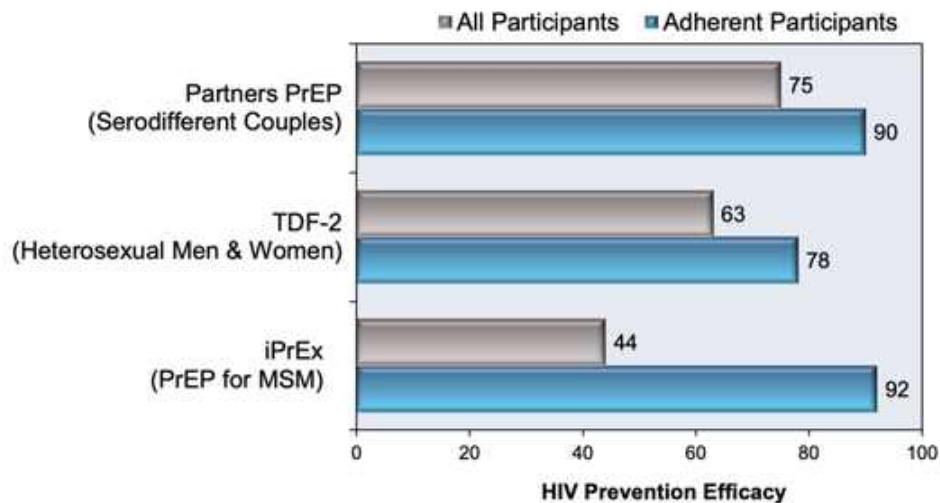


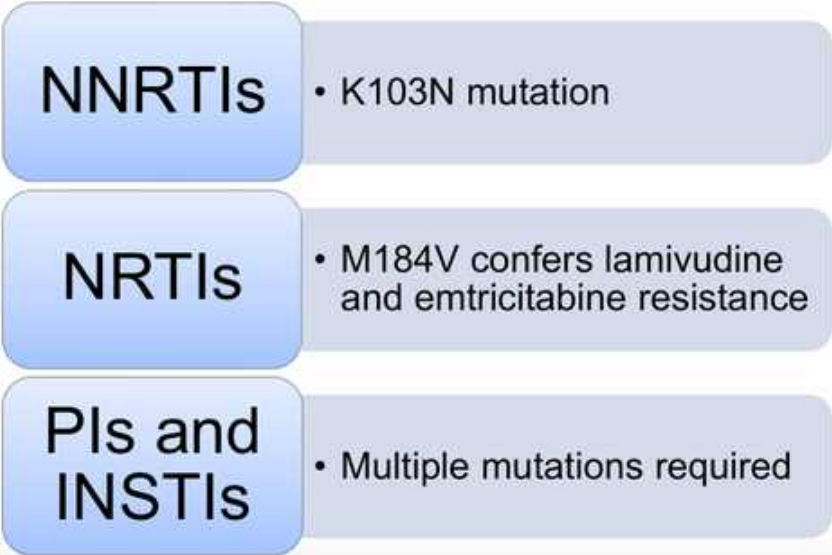
Image adapted from hiv.uw.edu

Slide 27. The PrEP efficacy figure visually reinforces that adherence strongly influences prevention efficacy.

Handout-only practical PEP detail: the companion specifies a **28-day, 3-drug ART regimen** started as soon as possible and no later than 72 hours after a significant exposure.

Slide 28: Resistance Patterns Worth Knowing

HIV Resistance



Slide 28. The deck gives three high-yield resistance rules rather than a giant mutation catalog.

Class	Deck resistance clue
NNRTI	K103N
NRTI	M184V → lamivudine + emtricitabine resistance
PI / INSTI	Multiple mutations generally required

RESISTANCE LOGIC

Adherence matters because ongoing replication creates opportunities for mutation. Suppression is not just a viral-load goal; it is also a resistance-prevention strategy.

Vignette Traps

Vignette	Answer
HLA-B*5701 positive	Avoid abacavir
Fanconi syndrome / bone-density loss	TDF
Vivid dreams / CNS symptoms	Efavirenz
Rilpivirine + aggressive acid suppression	Absorption problem / interaction
Polyvalent cations with ART	INSTI chelation
Cushing-like lipodystrophy + hyperglycemia	Protease inhibitor class
Need a tropism assay first	Maraviroc
New symptoms after ART reveals hidden infection	Unmasking IRIS
Needlestick after HIV exposure	PEP , not PrEP

FINAL 60-SECOND MAP

Entry: maraviroc / ibalizumab / fostemsavir • **RT:** NRTI + NNRTI • **Integration:** INSTI • **Maturation:** PI • **Capsid:** lenacapavir.

Then attach the big toxicities: **abacavir HLA-B*5701**, **TDF kidney/bone**, **NNRTI rash/SJS**, **INSTI cation chelation**, **PI metabolic/CYP**.

Source hierarchy: attached Anderson lecture deck is primary. The attached "Lecture 10" HIV Antiretroviral handout is secondary enrichment; current course numbering is Lecture 8 and the resource title is **HIV Antiviral Agents**.